Deno Jonas Feuerriegel

Mechanical Engineer · System Integration · Sensors · Hardware · Verification

Hamburg area | deno.feuerriegel@gmail.com | linkedin.com/in/deno-feuerriegel | denofeuerriegel.github.io

PROFESSIONAL EXPERIENCE

since 05/2025

Working Student, Product Development / Maritime Drone Technology

Projassment e.K.

- Independent development of an unmanned surface vehicle (USV) from the initial idea to a functional prototype
- Hardware development: design and integration of electronic components and custom PCBs; mechanical design of parts and assemblies
- Integration of sensors, propulsion and communication systems (UART); configuration of the autopilot software ArduPilot
- Build-up, commissioning and functional testing of prototypes; systematic troubleshooting and iterative optimisation of the overall system

03/2024 - 02/2025

Internship and Bachelor's Thesis in Powertrain Software Validation

Volkswagen AG

- Derivation, creation and execution of test cases and test plans from requirements for powertrain software and ECU functions on the Hardware-in-the-Loop test bench
- Bachelor's thesis: built an end-to-end, traceable verification chain from the requirement (PTC Codebeamer) through automated test execution (EXAM, HIL) to the documented test result: more than 200 test cases executable automatically
- Planned and independently carried out measurement drives with development and prototype vehicles; recorded and analysed ECU and sensor signals with ETAS INCA/MDA
- Evaluation of measurement data, comparison of parameter sets, documentation and analysis of deviations and abnormal software behaviour
- Created structured process and user documentation; onboarded colleagues into the test process

11/2021 - 02/2024

Working Student, Test Driver in Vehicle Testing

CoTec GmbH

- Test drives for the validation of advanced driver assistance systems under real-world conditions
- Identification and reproduction of functional anomalies; structured documentation of findings
- Technical alignment of test results with the engineering teams

EDUCATION

since 04/2025

M.Sc. Sustainable Energy Systems in Mechanical Engineering

HAW Hamburg (University of Applied Sciences)

Current grade average: 1.55 · expected graduation: 12/2026

- Methodological focus: systems engineering, development of mechatronic systems, statistical data analysis
- Master's thesis (ongoing): development and evaluation of a technical deployment system for seagrass restoration in the Baltic Sea

03/2021 - 03/2025

B.Sc. Mechanical Engineering and Production (Digital Engineering & Mobility)

HAW Hamburg (University of Applied Sciences)

Final grade: 1.5

- Bachelor's thesis at Volkswagen AG: automated software commissioning and validation on the HIL test bench
- Technical focus: mechatronics, automation, robotics, electric drive technology, finite element method

08/2017 - 01/2021

Apprenticeship as Automotive Mechatronics Technician

Kuhn + Witte GmbH & Co. KG

- Systematic fault diagnosis of electrical and electronic vehicle systems using diagnostic tools, measuring equipment, wiring diagrams and manufacturer information; bus systems (CAN, LIN, MOST, FlexRay)
- Maintenance and repair of mechanical systems; measurement and calibration of driver assistance sensors

2016

Abitur (university entrance qualification)

Gymnasium Winsen (Luhe)

SELECTED ACADEMIC PROJECTS

B.Sc.
Student project
Grade: 0.7

Digital PID Control Laboratory Experiment

Measurement and Control Engineering

- Design of a digital controller with Arduino Due; electronics developed with LTspice and KiCad
- Build-up, testing and documentation of the experiment; permanently adopted into the lab curriculum

M.Sc.
Team project
Grade: 0.7

Wind Farm Power Curve and Power Forecasting

CC4E wind farm Curslack

- Analysis and cleansing of operational and met-mast data; identification of faults and outages
- Power forecasts from live met-mast and weather data: $R^2 > 0.85$ and MAE approx. 500 kW at 12.6 MW rated capacity

SKILLS

Verification & Validation	Requirements analysis; test planning; test case design; test execution; traceability; deviation analysis; technical verification documentation
Test & Measurement	Hardware-in-the-Loop; test automation (EXAM); ETAS INCA/MDA; DiagRA; ODIS; vehicle testing
Requirements & Test Mgmt.	PTC Codebeamer; Atlassian Jira and Confluence; PTC Windchill
Hardware & Electronics	PCB design (KiCad); circuit simulation (LTspice); Arduino; sensor integration; UART; CAN/LIN
CAD/CAE	PTC Creo Parametric; Autodesk Fusion; ADINA/FEM
Programming & Data	Python; MATLAB/Simulink; Structured Text / PLC; Git; data analysis

LANGUAGES & INTERNATIONAL EXPERIENCE

Languages	German (native); English (business fluent)
Abroad	2017: three-month stay in the USA
Grading	German grading scale: 1.0 is the best grade (0.7 possible at HAW Hamburg), 4.0 is a pass